ChE 210 Homework Set #9 Due on Tuesday, 7 April 2020

(1) Murphy 5.40

(2) Murphy 5.41

(3) Air is heated from 25°C to 150°C prior to entering a combustion furnace. The change in specific enthalpy associated with this transition is 3640 J/mol. The flow rate of air at the heater outlet is 1.25 m^3/min and the air pressure at this point is 122 kPa absolute.

(a) Calculate the heat requirement in kW, assuming ideal gas behavior and that the kinetic and potential energy changes from the heater inlet to the outlet are negligible.

(b) Would the value of $\Delta \dot{E}_k$ [which was neglected in part (a)] be positive or negative, or would you need more information to answer? If you need more information, what would that information be?

(4) Murphy 6.26